

BRICKBUILDER ACCELERATOR · BILL OF MATERIALS

SQL → Databricks Migration Studio

A single, governed workspace that discovers, converts, migrates and validates legacy SQL Server, Snowflake, Redshift, SharePoint and REST API estates onto the Databricks Lakehouse — with metadata-driven pipelines and built-in data quality.

PRODUCT

Databricks Platform | Data Migration

PARTNER

Insight Enterprises

BUSINESS OBJECTIVE

Cut migration effort & risk by 60–70%

VERSION

v1.0 · August 2026

| Customer Challenge

Enterprises running mission-critical workloads on legacy SQL Server, Azure Synapse, Snowflake and Redshift face a hard deadline: aging infrastructure, rising licence costs, and the need for unified AI-ready governance on the Lakehouse. But migration programmes stall in manual, error-prone, hard-to-govern processes.

60%+

Manual Effort & Slow Delivery

Object-by-object conversion of stored procedures, views and UDFs to PySpark consumes engineer-months; schema and data movement are hand-built per table, with no repeatable pipeline factory.

Multi-Source

Fragmented Source Landscape

Estates span SQL Server, Azure SQL/Synapse, Snowflake, Redshift, SharePoint and REST APIs — each needing different connectors, discovery methods and validation logic, with no single control plane.

No Proof

Unproven Data Quality & Reconciliation

After cutover, teams struggle to prove completeness, accuracy and freshness to the business — quality checks are ad-hoc scripts, and reconciliation is manual spreadsheet work.

Accelerator Overview

A **single Databricks App** that walks a migration team through the full journey — **Discover** → **Convert** → **Deploy** → **Migrate** → **Validate** — on native Databricks services with Unity Catalog governance end-to-end.

1 Discovery & Complexity Engine

Live scans of source objects with automated complexity scoring, effort points, dependency graphs and BOM / HTML reports.

2 AI Convert to PySpark

LLM-powered conversion of stored procedures, views and UDFs to validated PySpark, with human-in-the-loop review.

3 MetadataFlow & Pipeline Studio

Configures metadata-driven Delta tables and auto-generates Extract / Bronze / Silver / Orchestrator notebooks — deploy once, run many.

4 Data Quality & Reconciliation

12 embedded DQ checks per pipeline run with quarantine, restore points, auto-rollback and **__dq_metrics** scoring; source-vs-target reconciliation reports.

5 Operations: Scheduler & Self-Healing

Job scheduling with watermark-based incremental loads, run history, and an automated health-check & self-healing bot.

6 Genie AI Assistant

Native chat over the migration workspace — asks questions of your metadata, pipelines and run history in plain English.

| Unique Technical Capabilities

A Metadata-Driven Pipeline Factory

Feature: Pipeline definitions live in Delta metadata tables (wf_pipeline_metadata, wf_job_metadata, wf_run_history). Generated notebooks read the metadata — one notebook serves every table.

Benefit: Onboard a new table in minutes, not days. Zero notebook sprawl; full audit trail in Unity Catalog.

B Built-in DQ Framework with Quarantine

Feature: 5 Bronze + 7 Silver checks — empty-file skip, null-key detection, dedup, schema-drift detection, quarantine flags, per-column null %, row-count anomaly alerts — scored per run into __dq_metrics, with restore points and auto-rollback.

Benefit: Bad rows never reach Silver silently. Every run ships with defensible, per-table quality evidence for cutover sign-off.

C Multi-Source, Multi-Cloud by Design

Feature: Pluggable connectors for Azure SQL, SQL Server, Synapse, Snowflake, Redshift, SharePoint (Entra ID app-only) and REST APIs with OpenAPI discovery — all configured from one Settings pane.

Benefit: One accelerator covers the whole estate; clients avoid buying point tools per source.

CUSTOMER OUTCOME

Migration teams move from spreadsheet-tracked, hand-coded projects to a governed factory: discovery-backed estimates, automated pipelines with embedded quality gates, and leadership dashboards showing live progress — typically eliminating the majority of manual build effort per table.

Scenario / Use Cases and Personas

Legacy SQL Estate Modernisation

- ✓ **Azure SQL / SQL Server / Synapse:** auto-discover objects, AI-convert T-SQL to PySpark, migrate via medallion pipelines.
- ✓ **Snowflake & Redshift:** connect with native credentials, inventory tables & objects, lift to Delta with reconciliation.
- ✓ **Assessment-first programmes:** complexity-scored BOM & effort estimates to scope and price the migration.

Unstructured & API Data Onboarding

- ✓ **SharePoint lists & libraries:** Entra app-only auth; lists surfaced as first-class migration tables.
- ✓ **REST APIs:** OpenAPI/Swagger auto-discovery; GET endpoints onboarded as governed Delta sources.
- ✓ **Landing zone:** UC Volumes staging with watermark-based incremental refresh.

Governed Run & Cutover Assurance

- ✓ **Quality-gated pipelines:** quarantine + DQ scores per run for sign-off evidence.
- ✓ **Reconciliation:** automated source-vs-target row/aggregate checks.
- ✓ **Operations:** schedules, watermarks, run history and self-healing for steady-state.

AUDIENCE · CIO / Data Platform Lead

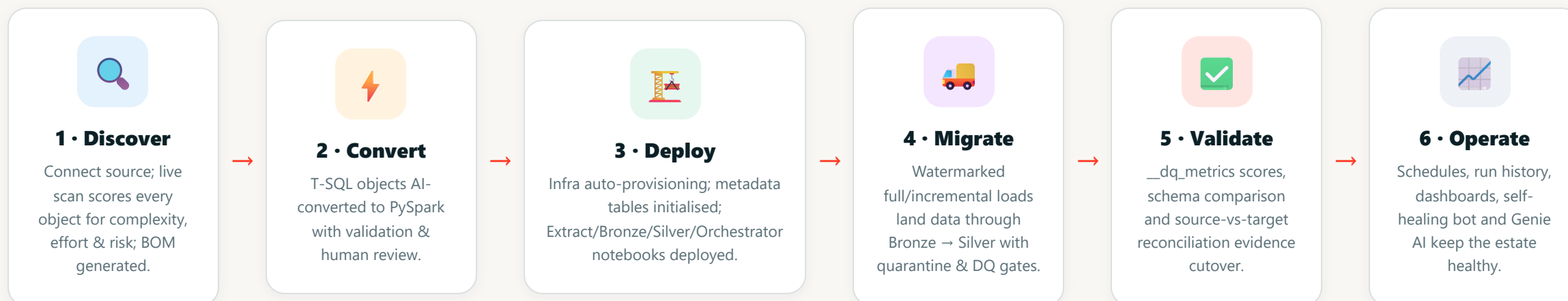
AUDIENCE · Migration Programme Manager

AUDIENCE · Data Engineer

AUDIENCE · Platform Architect

AUDIENCE · QA / Data Quality Lead

Process Flow

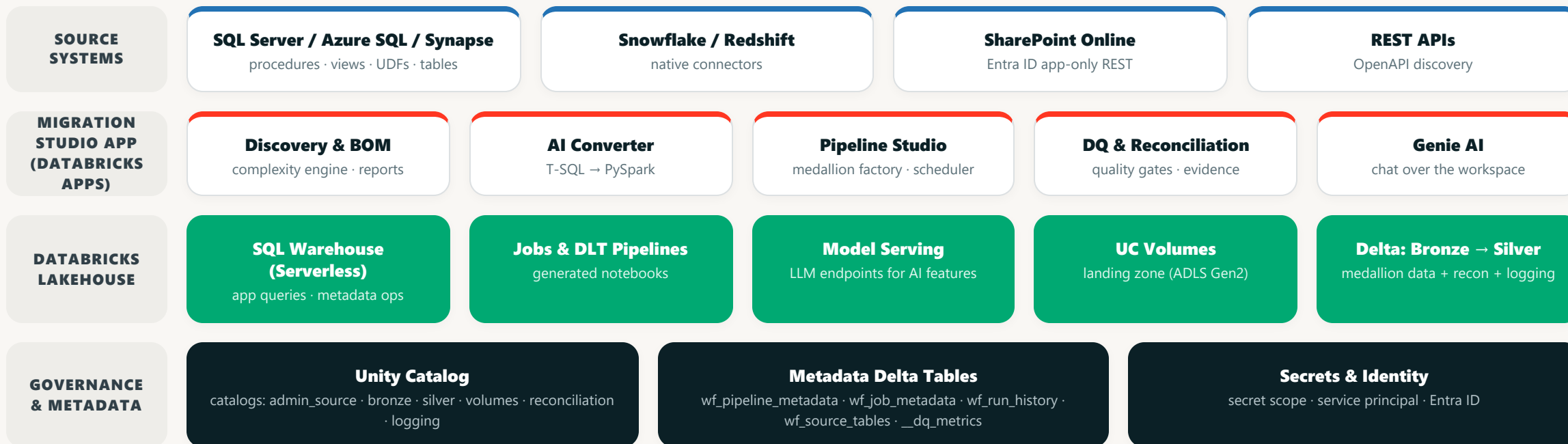


Orchestration — Databricks Workflows + Orchestrator notebook

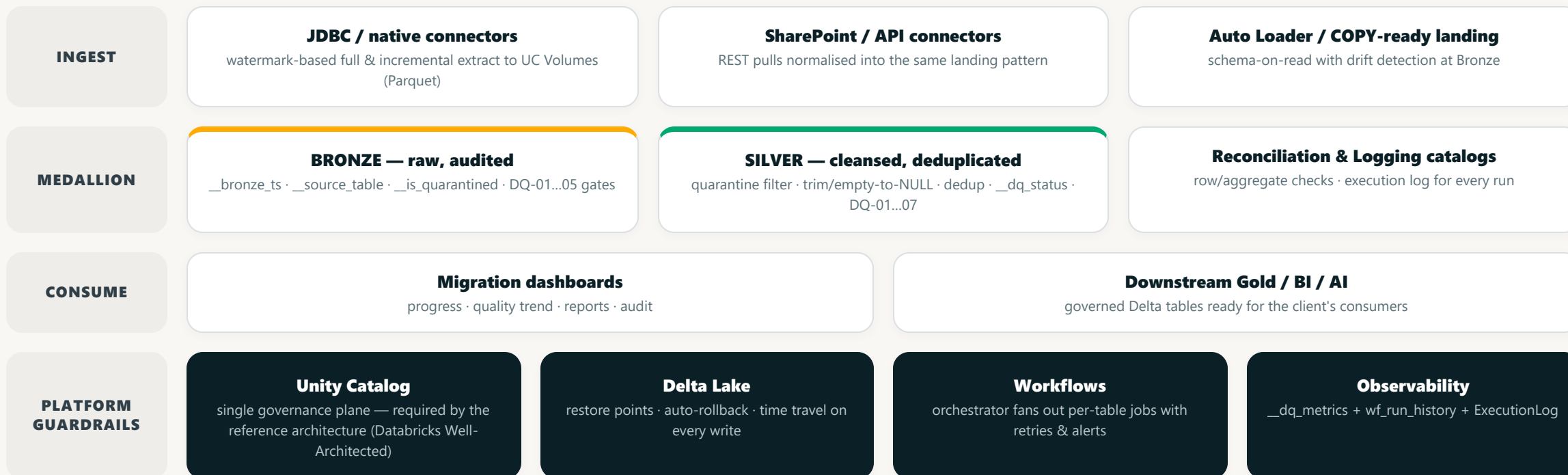
Governance — Unity Catalog on every catalog, schema & volume

Observability — wf_run_history + __dq_metrics + dashboards

Component Architecture



Accelerator Architecture



Bill of Materials — Deliverable Assets

ASSET	DESCRIPTION	FORMAT	AUDIENCE
Migration Studio App	Databricks Apps–hosted Flask application: Discovery, Convert, MetadataFlow, Pipeline Studio, Job Manager/Scheduler, Reports, Progress, Schema Compare, Reconciliation, Data Quality, Data Modeling, Health Check, Audit, Settings	Databricks App	Migration team
Metadata Notebooks ×4	01_Meta_Extract, 02_Meta_Bronze, 03_Meta_Silver, 00_Meta_Orchestrator — metadata-driven, DQ-gated, restore-point protected; DLT/SDP variants included	Notebooks + DLT	Platform / Data Eng
Data Quality Framework	12 checks (Bronze 5 / Silver 7), quarantine pattern, __dq_metrics scoring table, restore points & auto-rollback	Framework	QA / Data Governance
Source Connectors	Azure SQL / SQL Server / Synapse / Snowflake / Redshift / SharePoint (Entra ID) / REST API (OpenAPI discovery)	Modules	Data Eng
Analytics Dashboards	Progress funnel, quality trend & radar from __dq_metrics, reconciliation report, schema comparison, audit & compliance	Dashboards	PMO / Leadership
Self-Healing Bot	Automated health checks with healing rules and restore-point recovery	Automation	Operations
Genie AI Integration	Natural-language assistant over workspace metadata & run history via Model Serving endpoints	AI Feature	All users
Enablement Pack	Deployment runbook, interactive Prerequisites & Go-Live checklist, admin/user guide, architecture deck	Docs	Client team

Solution Description

Solution Summary

Deployment of Migration Studio into the client's Azure Databricks workspace, configuration against up to two source systems, pilot migration of a representative table group through the full DQ-gated medallion flow, and handover with enablement.

- ✓ App deployment + environment configuration (catalogs, warehouse, secrets, storage)
- ✓ Discovery assessment with complexity-scored BOM for the pilot scope
- ✓ Pilot: source → Bronze → Silver with DQ metrics & reconciliation evidence
- ✓ Hypercare + admin/developer enablement sessions

2–4 wks

Typical timeline — deploy to validated pilot handover (per source system pair)

60–70%

Manual effort reduction vs hand-coded object-by-object migration

This Solution Unlocks

- ✓ Board-level, live view of migration progress & quality
- ✓ Repeatable factory for every remaining wave
- ✓ Governed, AI-ready Delta estate on Unity Catalog

Contact Us



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Let's Partner — bring Migration Studio to your next SQL modernisation pursuit.